

PAVEL ZOSIM

TECHNICAL ARTIST / PROCEDURAL SYSTEMS / REAL-TIME GRAPHICS

HOUDINI · UNREAL · UNITY · VEX · HLSL · PYTHON · GPU WORKFLOWS

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PROFILE_01 PROFILE

Technical Artist with 6+ years in real-time production, focused on Houdini-driven procedural systems, shaders, VFX, and engine-facing workflows. Graphics-first by background: I use geometry, points, attributes, simulation, and data-driven rules as the common language across Houdini, Unreal, Unity, and production tools. I use Python, C#, VEX, and HLSL pragmatically to implement visual systems, procedural workflows, and production tooling.

Production work spans Meta / Blumhouse XR, NASA First Woman, interactive installations, procedural environments, and mobile real-time VFX. I tend to work between art and engineering: reducing repetitive setup, making procedural systems predictable, and keeping artist control explicit.

CAPABILITIES_02 CORE CAPABILITIES

PROCEDURAL

Houdini HDAs · VEX · procedural geometry · deterministic generation · PDG/TOPs · USD/Solaris

REAL-TIME

Unreal Engine 5 · Unity · Niagara · VFX Graph · HISM/instancing · VR/XR optimization

SHADERS / GPU

HLSL · Compute Shaders · Unreal Materials · Shader Graph · GPU-driven effects · overdraw / frame-time profiling

TOOLS

Python · C# editor tools · exporters · validation · data-driven setup · documentation

FX / SIM

Pyro · RBD · FLIP · VDB · VAT · vector fields · flipbooks · Six Point Lighting

PIPELINES

Asset handoff · MaterialX · ACES · Arnold / Karma / Redshift / Octane · Git / Perforce

CONTENT

Substance Designer / Painter · SpeedTree · ZBrush · procedural environment workflows

COMPOSITING

AOV workflows · Cryptomatte · After Effects · working knowledge of Nuke and colour grading

SYSTEMS_03 SELECTED SYSTEMS

SYS_A Procedural Authoring Architecture – Houdini → Unreal

Modular level and environment framework with deterministic generation, manual overrides, gameplay placement, asset registries, and separate routing for repeated structural instances versus Blueprint actors that require logic.

Houdini HDA · VEX/Python · Unreal Engine · HISM · Blueprint

SYS_B Runtime Representation & Placement Routing – HISM / Blueprint

Built a semantic placement layer on top of Houdini Engine: Houdini assigns representation and target metadata; Unreal owns meshes, materials, collision, and components. Replaced thousands of structural Blueprint actors with named HISM batches while preserving actor-based gameplay modules.

Houdini Engine · Unreal integration · HISM · Blueprint · data contracts

SYS_C Procedural FX Export System

Layer-based Houdini explosion architecture with deterministic pre-sim, Pyro sourcing, and multiple outputs from one setup: VDB for offline rendering, VAT / Six Point Lighting for real-time playback, vector fields, and mesh cache.

Houdini · Pyro · VDB · VAT · Six Point Lighting · Unity/Unreal

SYS_D Production Utilities & Bridges

Published tools include Houdini → After Effects data transfer, procedural rig/skeleton generation, folder-based asset import, material path management, cubemap assembly, and sprite-sheet assembly. Tools are documented and sold externally through Gumroad.

Python · VEX · Houdini · pywebview/Pillow · After Effects scripting

PRODUCTION_04 PRODUCTION EXPERIENCE

TECHNICAL ARTIST / VFX DEVELOPER

Freelance Contract · Apr 2024 - Present

- Meta / Blumhouse XR – developed custom HLSL shaders and real-time VFX for standalone Meta Quest 3 / 3S; prototyped effect workflows in Houdini before Unity integration.
- Reduced GPU frame time by approximately 2-4 ms through shader simplification and overdraw reduction while balancing real-time effects with streamed 4K video and spatial audio.
- For a live Unity match-3 project, redesigned a duplicated VFX setup around shared parameter-driven SubGraphs and a root controller: asset count reduced from roughly 90 files to about 40, with shared fixes propagated from one base setup.
- Documented integration, debugging, naming, folder, prefab, shader, and flipbook conventions for handoff and maintenance.

TECHNICAL ARTIST / VFX ARTIST / MOTION DESIGNER

Bully! Entertainment · Jun 2018 - Apr 2024

Small-studio role spanning concept, UI/UX, 3D, motion, shaders, Houdini tooling, and final real-time delivery; frequently the bridge between art and engineering.

NASA / FIRST WOMAN

- Owned visual and technical decisions across app, web, graphic-novel, and video outputs; built Houdini-driven screen/UV tooling and shaders for multi-format delivery.
- Automated unified texture-atlas / UV generation across screen variants, eliminating per-variant manual UV work and reducing material overhead.
- Project reach included 1.5M website page views, 325K+ app downloads, and 350K+ graphic-novel copies distributed; astronaut Frank Rubio read the content aboard the ISS.

DELOITTE / INTERACTIVE & AR

- Built texture export and sprite-sheet tooling used across production, reducing manual asset-preparation overhead by approximately 30-40% per cycle.
- Created scalable UI/material workflows for interactive and AR installations, including atlases, 9-slice assets, screen/world-space UI, and real-time effects.

FAA / AIRPORT DIGITAL TWIN R&D

- Generated city and airfield content procedurally with Houdini + OpenStreetMap + Python, replacing large amounts of manual modeling and enabling rapid iteration.
- Built Unity HDRP timeline-driven aircraft animation and touch-camera workflows for an installation designed for continuous long-duration operation.

FOUNDATION_05 EARLY CAREER FOUNDATION

UI/UX, MOTION & 3D DESIGN

2005 - 2018

Thirteen years across app/web UI, motion graphics, 3D, print, photography, and post-production. This visual background is the basis for later technical work: composition, timing, colour, asset structure, and the habit of automating repetitive production tasks.

BACKGROUND_06 EDUCATION & LANGUAGES

ULIM – Universitatea Liberă Internațională din Moldova, Chisinau

Computer Science / CG Design studies – incomplete

Self-directed technical development: Houdini, procedural graphics, real-time rendering, VEX/Python, Unity/Unreal, USD, shaders, simulation, and production tooling.

Languages: Russian – Native · English – Professional Working Proficiency

PUBLIC_07 SELECTED PUBLIC WORK

BREAKDOWNS

[Controlled Procedurality](#)
[Houdini → Unreal Typed Placement Routing](#)
[Houdini Procedural Explosion VFX](#)

PIPELINE NOTES

[Houdini → After Effects Data Export](#)
[Procedural Interiors with Houdini & USD](#)
[procedural environment studies](#)

COMMERCIAL TOOLS

[Houdini → After Effects Bridge](#)
[Procedural Skeleton Generator](#)
[Folder-Based File Importer](#)
[Material Path Manager](#)

READ / WATCH

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